AI MOVIE RECOMMENDATION SYSTEM

CODE.PY > ...

```python
# ==============================================
# AI Recommendation System
# Project 3 - DecodeLabs
# ==============================================

# Dataset of movies with their genres

movies = [
    {"name": "Inception", "genres": ["Action", "Sci-Fi", "Thriller"]},
    {"name": "Interstellar", "genres": ["Sci-Fi", "Adventure", "Drama"]},
    {"name": "The Dark Knight", "genres": ["Action", "Crime", "Drama"]},
    {"name": "Titanic", "genres": ["Romance", "Drama"]},
    {"name": "Avengers: Endgame", "genres": ["Action", "Adventure", "Sci-Fi"]},
    {"name": "The Conjuring", "genres": ["Horror", "Thriller"]},
    {"name": "Frozen", "genres": ["Animation", "Family", "Adventure"]},
    {"name": "Toy Story", "genres": ["Animation", "Comedy", "Family"]},
    {"name": "3 Idiots", "genres": ["Comedy", "Drama"]},
    {"name": "Bahubali", "genres": ["Action", "Adventure", "Drama"]},
    {"name": "Doctor Strange", "genres": ["Action", "Fantasy", "Sci-Fi"]},
    {"name": "Spider-Man: No Way Home", "genres": ["Action", "Adventure", "Fantasy"]},
    {"name": "Joker", "genres": ["Crime", "Drama", "Thriller"]},
    {"name": "The Notebook", "genres": ["Romance", "Drama"]},
    {"name": "Finding Nemo", "genres": ["Animation", "Adventure", "Family"]}
]

# ==============================================
# Display Available Genres
# ==============================================

available_genres = sorted(set(
    genre for movie in movies for genre in movie["genres"]
))

print("=" * 50)
print("        AI MOVIE RECOMMENDATION SYSTEM")
print("=" * 50)
```

```python

print("\nAvailable Genres:")
for genre in available_genres:
    print("-", genre)

# ==============================================
# Take User Input
# ==============================================

user_input = input(
    "\nEnter your favorite genres (comma separated): "
)

user_preferences = [
    genre.strip().title()
    for genre in user_input.split(",")
]

# ==============================================
# Recommendation Logic
# ==============================================

recommendations = []

for movie in movies:

    score = len(
        set(user_preferences) &
        set(movie["genres"])
    )

    if score > 0:
        recommendations.append((movie["name"], score, movie["genres"]))

# ==============================================
# Sort Recommendations
# ==============================================
```

C:\Users\vemul\OneDrive\Desktop\decodelabs.internship\CODE.PY

```python
recommendations.sort(key=lambda x: x[1], reverse=True)

# ==============================================
# Display Results
# ==============================================

print("\n" + "=" * 50)

if recommendations:

    print("Recommended Movies\n")

    for movie, score, genres in recommendations:
        print(f"Movie : {movie}")
        print(f"Genres: {', '.join(genres)}")
        print(f"Similarity Score: {score}")
        print("-" * 50)

else:
    print("No matching recommendations found.")

print("\nThank you for using the AI Recommendation System!")
```

OUTPUT

```
PS C:\Users\vemul\OneDrive\Desktop\decodelabs.internship> & C:/Users/vemul/AppData/Local/Programs/Pyth
eDrive/Desktop/decodelabs.internship/task 2.py"
First Five Rows
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

Accuracy: 1.0

Classification Report
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        10
           1       1.00      1.00      1.00         9
           2       1.00      1.00      1.00        11

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```

```
PS C:\Users\vemul\OneDrive\Desktop\decodelabs.internship> & C:/Users/vemul/AppData/Local/Prog
Drive/Desktop/decodelabs.internship/CODE.PY
==================================================
        AI MOVIE RECOMMENDATION SYSTEM
==================================================

Available Genres:
- Action
- Adventure
- Animation
- Comedy
- Crime
- Drama
- Family
- Fantasy
- Horror
- Romance
- Sci-Fi
- Thriller

Enter your favorite genres (comma separated): Traceback (most recent call last):
  File "c:\Users\vemul\OneDrive\Desktop\decodelabs.internship\CODE.PY", line 46, in <module>
    user_input = input(
        "\nEnter your favorite genres (comma separated): "
    )
KeyboardInterrupt
PS C:\Users\vemul\OneDrive\Desktop\decodelabs.internship>
```